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|--------------|-----------------------------------|
| Started on   | Sunday, 12 February 2023, 8:45 PM |
| State        | Finished                          |
| Completed on | Sunday, 12 February 2023, 8:45 PM |
| Time taken   | 14 secs                           |
| Marks        | 0/17                              |
| Grade        | 0 out of 100                      |

## Question 1

Not answered

Marked out of 1

If anhedonia scores as measured by Beck's depression inventory are found to correlate with treatment response to depression as measured by changes in Montgomery Asberg scale (Pearson's coefficient 'r' of 0.88,  $p < 0.05$ ), which of the following is true?

Select one:

- ☐ Treatment responsivity is not related to degree of anhedonia
- ☐ Lesser the anhedonia, better the treatment response.
- ☐ Higher the anhedonia, greater the treatment response
- ☐ Treatment responsivity causes anhedonia
- ☐ Anhedonia causes treatment response

Your answer is incorrect.

Examining the correlation coefficient, it is clear that it is close to 1. Hence, a positive linear relationship exists between the two variables. We cannot conclude causal relationship as correlation does not always imply causation.

The correct answer is: Higher the anhedonia, greater the treatment response

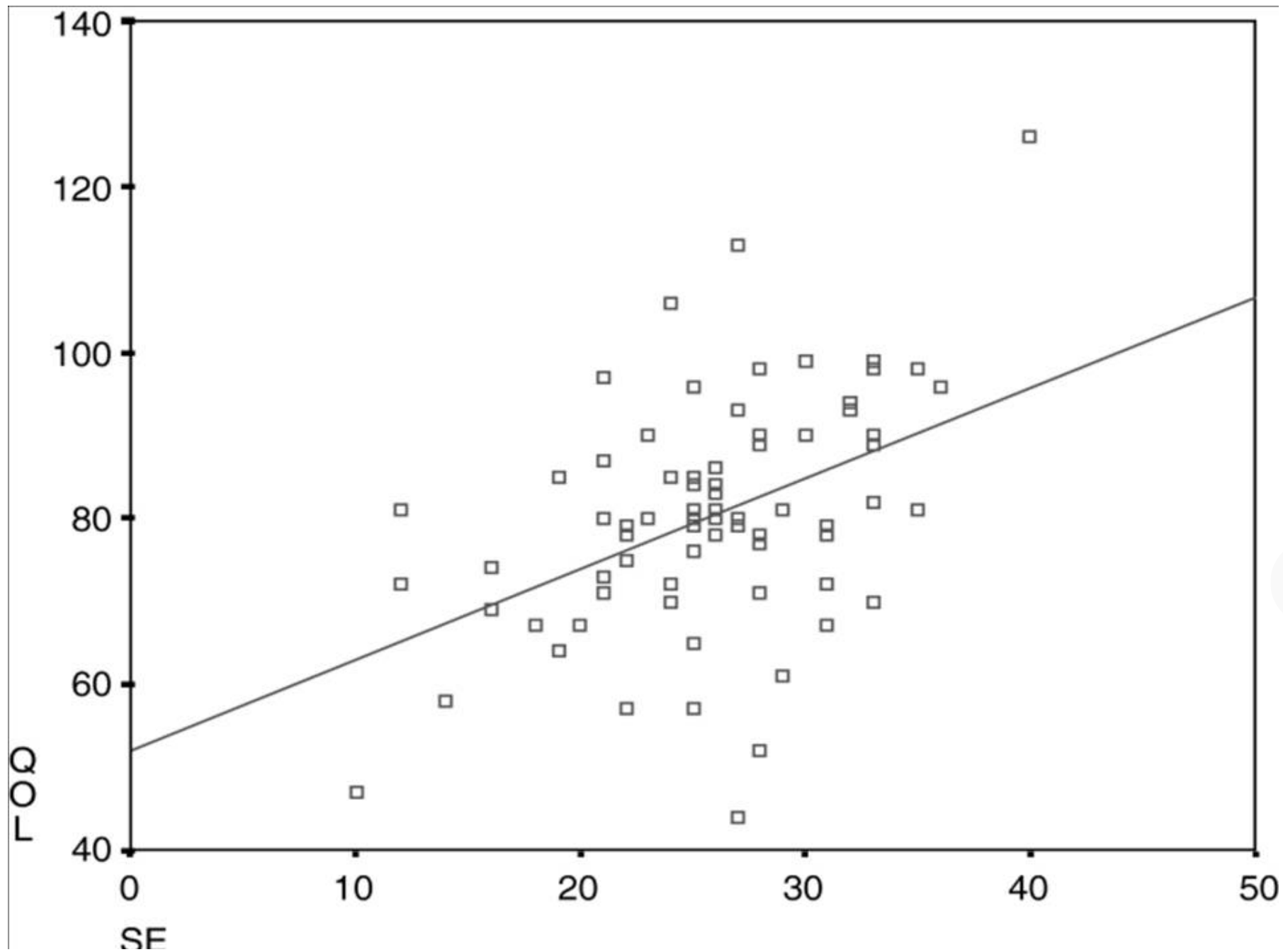
Question **2**

Not answered

Marked out of 1

The below image shows a scatter plot of perceive the quality of life (QOL) and self-esteem (SE) in patients living in community with schizophrenia. Which of the following inferences is NOT correct?





Select one:

- ☐ Quality of life increases as self esteem increases
- ☐ It is likely that the correlation coefficient is closer to one than to zero
- ☐ The association shows a positive slope
- ☐ The correlation coefficient is 0.75
- ☐ Causal association cannot be determined from this graph

Your answer is incorrect.

A somewhat positive linear relationship can be inferred from the scattergram. As a scatterplot is not quantitative, it is not possible to infer the magnitude of the relationship between the two variables. Kunikata, Y. M. K. N. (2005) Quality of life of schizophrenic patients living in the community: The relationships with personal characteristics, objective indicators, and self-esteem. *Psychiatry and Clinical Neurosciences*, 59, 163-169.

The correct answer is: The correlation coefficient is 0.75



## Question 3

Not answered

Marked out of 1

The range of correlation coefficient is usually between

Select one:

- ☐ (0) and (100)
- ☐ (1) and (10)
- ☐ (-1) and (+1)
- ☐ (-1) and (0)
- ☐ (0) and (+1)

Your answer is incorrect.

The correlation coefficient  $r$  must take on a value from -1 to 1. A value of  $r = 1$  suggests a perfect positive correlation in which the association between  $x$  and  $y$  can be described by a straight line with a positive slope. A value of  $r = -1$  indicates a perfect negative correlation.

The correct answer is: (-1) and (+1)



Question **4**

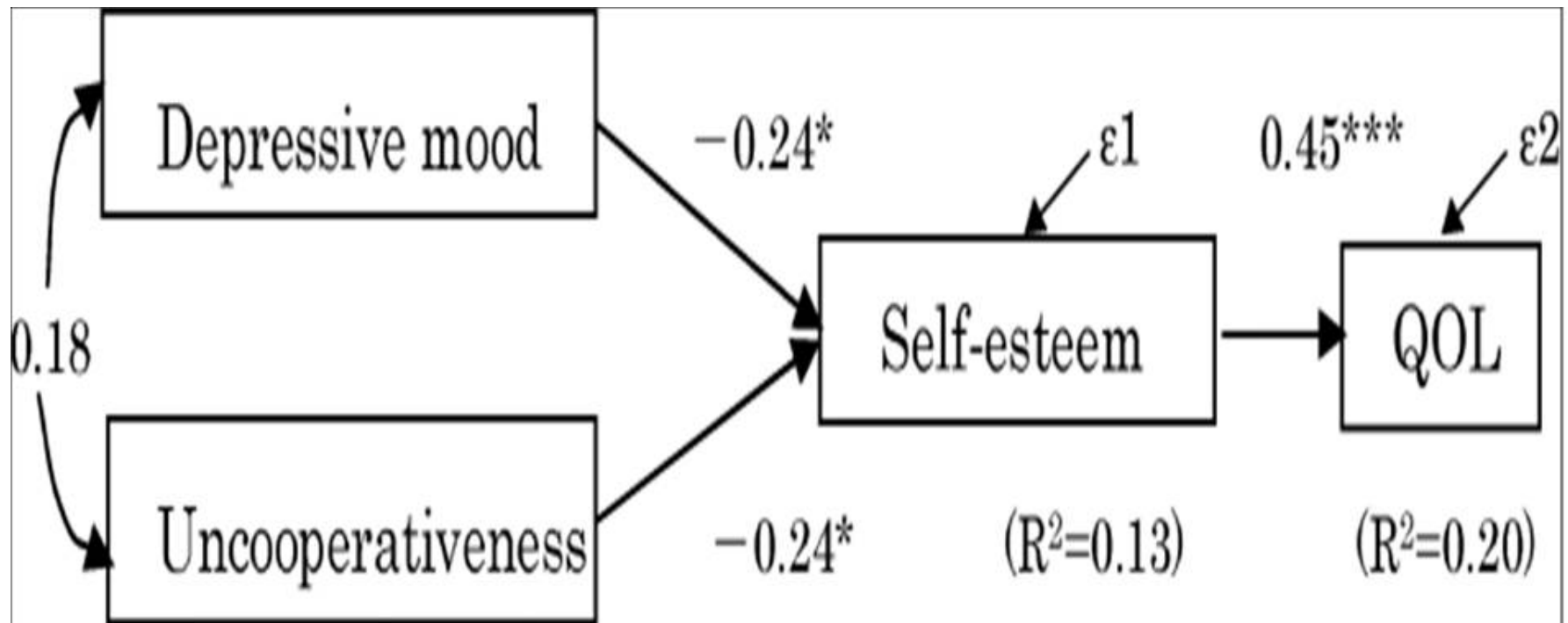
Not answered

Marked out of 1



This figure represents which of the following statistical analyses







Select one:

- ☐ Meta analysis
- ☐ Path analysis
- ☐ Root cause analysis
- ☐ Survival analysis
- ☐ Economic analysis

Your answer is incorrect.

Path analysis is an extension of multiple regression. It goes beyond regression in that it allows for the analysis of more complicated models. In particular, it can examine situations in which there are several final dependent variables and those in which there are chains" of influence, in that variable A influences variable B, which in turn affects variable C. Despite its previous name of "causal modelling," path analysis cannot be used to establish causality or even to determine whether a specific model is correct; it can only determine whether the data are consistent with the model". Retrieved from Finding our way: an introduction to path analysis. (n.d.). Retrieved from <http://www.ncbi.nlm.nih.gov/pubmed/15807228> - Can J Psychiatry 2005;50:115-122

The correct answer is: Path analysis



## Question 5

Not answered

Marked out of 1

Consider a linear regression equation be  $y = ax_1 + bx_2 + c$ . Which of the following denotes the dependent variable in this equation?

Select one:

- ☐ y
- ☐ a
- ☐ c
- ☐ b
- ☐  $x_1$

Your answer is incorrect.

Y is the dependent variable. It depends on values of  $x_1$  and  $x_2$ , which are independent variables.

The correct answer is: y



## Question 6

Not answered

Marked out of 1

To construct a multiple regression equation, which of the following criteria must be met?

Select one:

- ☐ The dependent variable must be categorical
- ☐ The independent variable must be continuous
- ☐ The dependent variable must be continuous
- ☐ The independent variable must be categorical
- ☐ The dependent variable must be dichotomous

Your answer is incorrect.

To construct a regression analysis, the dependent variable must be continuous and must be normally distributed. For correlation, both the dependent variable and the independent variable must be normally distributed. For regression analysis, only the dependent variable need be normally distributed. The independent variable does not need to be normally distributed;

The correct answer is: The dependent variable must be continuous

## Question 7

Not answered

Marked out of 1

Online workplace based peer appraisals for psychiatric trainees have two components. A self-appraisal form must be completed by each trainee rating him/herself in 6 different domains. This is compared with a composite score given by peer group in the same domains. The most appropriate statistical test for the comparison is

Select one:

- ☐ Student t test
- ☐ Trainee t test
- ☐ Chi-square test
- ☐ Kappa statistics test
- ☐ Correlation analysis

Your answer is incorrect.

For data that are not continuous measurements, the statistical method most commonly associated with reliability and agreement is called the kappa coefficient. Here agreement of scores in various domains is tested for a categorical outcome.

The correct answer is: Kappa statistics test



## Question 8

Not answered

Marked out of 1

The Pearson correlation coefficient between variables A and B is known to be -0.50, and the correlation between B and C is known to be 0.50. What can we infer about the relationship between A and C

Select one:

- ☐ A and C are independent.
- ☐ There is no association between A and C.
- ☐ A and C have a linear association.
- ☐ The relationship cannot be determined from the information given
- ☐ A and C are negatively correlated.

Your answer is incorrect.

The relationship cannot be determined. The observed values of A while plotting against B may be different from the observed values of A while plotting against C. Hence we cannot compare these two different correlations.

The correct answer is: The relationship cannot be determined from the information given



## Question 9

Not answered

Marked out of 1

An investigator studies 50 pairs of mixed-sex twins and reports the following equation that gives the regression of female birth weight (y) on male birth weight (x) (all weights in grams):  $y = 1200 + 0.4x$ . Which of the following is a valid conclusion from this?

Select one:

- ☐ Mean weight of twin brothers of girls who weigh 1000 g is predicted to be 1600 g.
- ☐ The sample correlation between girl's weight and boy's weight is 1200
- ☐ Mean weight of twin sisters of boys who weigh 1000 g is predicted to be 1600 g
- ☐ The sample mean weight of the boys is 1200 g
- ☐ The sample mean weight of the girls is 1200 g

Your answer is incorrect.

Here the female birth weight (y) is the dependent variable. Hence, we cannot predict male birth weight from this equation. The correlation coefficient is not given by this equation -the regression coefficient is 0.4. For a value of 1000 (x), the value of (y) is  $1200 + 1000(0.4) = 1600$ .

The correct answer is: Mean weight of twin sisters of boys who weigh 1000 g is predicted to be 1600 g



## Question 10

Not answered

Marked out of 1

Which of the following concerning correlation coefficient is incorrect?

Select one:

- ☐ A positive value implies that a rise in one variable accompanies rise in the other
- ☐ Is demonstrated by the symbol kappa
- ☐ Describes the degree of association between two variables
- ☐ Spearman's correlation coefficient is a non-parametric method of detecting correlation coefficient
- ☐ It is a measure on a scale of -1 to 1

Your answer is incorrect.

The correlation coefficient is given by Spearman's 'rho' (nonparametric) or Pearson's 'r' (parametric). It is expressed in a scale between -1 to +1. A positive value indicates that a rise in one value is associated with a rise in the other.

The correct answer is: Is demonstrated by the symbol kappa



## Question 11

Not answered

Marked out of 1

The best way to recognise whether or not a variable is growing exponentially over time is by:

Select one:

- ☐ Calculating a mean value across the different time points
- ☐ Calculating the least squares regression line of the variable against time and examining residuals
- ☐ Plotting variable against time and looking for a straight-line pattern
- ☐ Plotting logarithm of a variable against time and looking for a straight line pattern
- ☐ Using Cohen's kappa measurement

Your answer is incorrect.

If one wants to demonstrate the exponential relationship of a variable with a factor such as time, log-transformed values can be plotted against time. This is called exponential correlation.

The correct answer is: Plotting logarithm of a variable against time and looking for a straight line pattern





## Question 12

Not answered

Marked out of 1

The goodness of fit between two or more variables can be tested using which of the following statistical methods?

Select one:

- ☐ Analysis of variance
- ☐ Unpaired t test
- ☐ Correlation
- ☐ Student's t test
- ☐ Regression

Your answer is incorrect.

R<sup>2</sup> or the coefficient of determination is used to test the goodness of fit or final regression. It measures how well actual outcome (dependent variable) and calculated dependent variable correspond with each other.

The correct answer is: Regression



## Question 13

Not answered

Marked out of 1

Which of the following methods starts with most significant to least significant independent variable according to the correlation coefficient values when constructing a multiple regression equation?

Select one:

- ☐ Logistic regression
- ☐ Central regression
- ☐ Forward selection
- ☐ Backward elimination
- ☐ Stepwise regression

Your answer is incorrect.

Three methods are described for performing regression: 1. Stepwise regression: Calculates coefficient of regression and starts with most significant to least significant independent variable and fits them in a stepwise fashion into regression equation. But a disadvantage is sometimes the statistically significant variables may not be clinically significant. 2. Forward selection: A confounding factor by definition is associated with both independent variable and dependent variable. As one may not know which is a confounding variable in some occasions (this is what one often examines with multiple regression), these are treated as covariates. While constructing a multiple regression equation, if the regression coefficient of a previously added variable changes then either one of the covariates is a confounder; so these are retained in the equation irrespective of statistical significance. The latter added covariate is discarded if no change occurs in regression coefficients. 3. Backward elimination: This starts with the final model - the full equation and tries to discard covariates one by one according to changes that occur in correlation coefficients.

The correct answer is: Stepwise regression

## Question 14

Not answered

Marked out of 1

Agreement in diagnosis between GPs and psychiatrists is being tested. All of the following measures are needed to calculate the agreement except

Select one:

- ☐ Actual disagreement beyond chance
- ☐ Number of occasions when both groups agree with presence of diagnosis
- ☐ Number of occasions when both groups agree with absence of diagnosis
- ☐ Agreement expected on the basis of chance
- ☐ Actual agreement beyond chance

Your answer is incorrect.

Association refers to how much two measures 'go together'; agreement refers to the 'degree to which they are actually the same'. Suppose two instruments are used to diagnose the presence of depression. The scales achieve same answer while questioning about some symptoms of depression but not on others. The agreement or concordance can be simply measured as the (number of commonly agreed symptoms/ total number of symptoms). But this does not take into account that some agreement may be due to chance itself. Cohen's kappa allows for such a chance agreement while calculating actual agreement (beyond chance). It is expressed as a value between 0 and 1. It is sensitive to baseline prevalence rate.  $\text{Kappa} = \frac{\text{Observed agreement beyond chance}}{\text{Potential agreement beyond chance}}$

The correct answer is: Actual disagreement beyond chance

## Question 15

Not answered

Marked out of 1

A well-established depression scale (A) with 100 items and a newly designed scale (B) with 12 items are compared. Scale B takes 10 minutes to administer while an hour-long interview is needed for scale A. Both scales A and B were administered to 100 depressed in-patients and 100 controls. Spearman's correlation  $r = 0.8$  with  $p < 0.01$  was obtained when scores are compared. High scores on A correlated with high scores on B. Which of the following assertion can be made from the above findings?

Select one:

- ☐ A is a good screening tool
- ☐ A and B show good inter-rater reliability
- ☐ B can replace A
- ☐ A and B are highly LINEARLY correlated
- ☐ B is a good measure of the severity of depression

Your answer is incorrect.

The significant positive correlation factor indicates that scale A and B are highly positively correlated. One cannot decide on which is a better clinical tool using the above data.

The correct answer is: A and B are highly LINEARLY correlated



## Question 16

Not answered

Marked out of 1

A well-established depression scale (A) with 100 items and a newly designed scale (B) with 12 items are compared. Scale B takes 10 minutes to administer while an hour-long interview is needed for scale A. Both scales A and B were administered to 100 depressed in-patients and 100 controls. Spearman's correlation  $r = 0.8$  with  $p < 0.01$  was obtained when scores are compared. High scores on A correlated with high scores on B. Using the above study indicate which of the following statements is correct:

Select one:

- ☐ A is more valid than B
- ☐ Because A is longer than B it must be better
- ☐ B is more valid than A
- ☐ B is more reliable than A
- ☐ No comments could be made on validity/reliability

Your answer is incorrect.

The significant positive correlation factor indicates that scale A and B are highly positively correlated. One cannot comment on validity or reliability using a head to head comparison of 2 different scales. Longer scales need not be necessarily superior to shorter ones.

The correct answer is: No comments could be made on validity/reliability

## Question 17

Not answered

Marked out of 1

|                                     | Hypomania<br>present (Research<br>Psychiatrist) | Hypomania absent<br>(Research<br>Psychiatrist) | Total |
|-------------------------------------|-------------------------------------------------|------------------------------------------------|-------|
| Hypomania<br>present<br>(Clinician) | 41                                              | 3                                              | 44    |
| Hypomania<br>absent<br>(Clinician)  | 4                                               | 27                                             | 31    |
|                                     | 45                                              | 30                                             | 75    |

What is the observed agreement in the above table?

Select one:

- ☐ 0.907  
☐ 0.177  
☐ 0.517

- ☐ 0.413
- ☐ 0.81

Your answer is incorrect.

The observed agreement is not the same as Kappa. Here the two raters agree 41 times for the presence of hypomania and 27 times for the absence of hypomania. Totally they agree 68 times out of possible 75 instances. The agreement rate is  $68/75 = \text{nearly } 90\%$

The correct answer is: 0.907



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